3.5 METHODS OF ANALYSIS:

The structural analysis is a mathematical algorithm process by which the response of a structure to specified loads and actions is determined. This response is measured by determining the internal forces or stress resultants and displacements or deformations throughout the structure.

To get various forces and moments in the members (or) frames, it is necessary to analyze the frame and then designed. The following are some methods used in frame analysis.

1. Slope Deflecting Method.
2. Moment Distribution Method.
3. Kani's Method.
4. Substitute Frame Method.
5. Finite Element Method.
6. Cantilever Method.
7. Portal Method.
8. Factor Method.
9. Matrix Method.
10. Column Analogy Method.

In this project Substitute Frame Method of Analysis has been adopted, since it is easier and less time-consuming manual process.

3.5.1 Kani's Method:

Gasper Kani, a German engineer, developed another distribution procedure based on slope deflection equations. This method is very useful for the analysis of multistorey frames. The greatest advantage of this method is, even if a mistake is committed in distribution in one of the cycles, it converges finally to the correct answer. Even today, many practising engineers who are not familiar with computer methods, use Kani's method for the analysis of 3 to 4 storey building frames.

This method is first explained for structures with fixed ends. Then, the modifications to handle simply supported and overhanging ends are discussed. Analysis of symmetric frames making use of the symmetry is also explained, after which an analysis of general frames is taken up.